

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical
Practical – VI

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Class: TYIT	Batch:01
Date of Assignment: 30-07-2026	Date/Time of Submission: 30-07-2026

- a. **Create a simple website for a College Information Portal and convert it into a Progressive Web Application.**

Code:

#Create wwwroot/manifest.json:-

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icon/icon2.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icon/icon3.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

#Create wwwroot/service-worker.js:-

```
self.addEventListener("install", e => {
  console.log("Service workers Installed");
});
self.addEventListener("fetch", e => {
  e.responseWith(fetch(e.request));
});
```

#Add to _Layout.cshtml

#Inside the <head> section:

```
<link rel="manifest" href="/manifest.json"/>
```

#Before the closing </body> tag:

```
<script>
  if('serviceWorker' in navigator){
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>
```

Output:-

Welcome

Learn about [building Web apps with ASP.NET Core](#).



Install HTML TO PWA app

Publisher: localhost:7198

Use this site often? Install the app which:

- Opens in a focused window
- Has quick access options like pin to taskbar
- Syncs across multiple devices

Install

Not now

- b. Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:

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    },
    {
      "src": "icon/icon3.png",
      "sizes": "447X447",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

#Create wwwroot/service-worker.js:-

```
const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offline.html"
      ]);
    })
  );
  console.log("Service Worker Installed");
});
self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() => caches.match("/offline.html"))
  );
});
```

#Add to _Layout.cshtml

#Inside the <head> section:

```
<link rel="manifest" href="/manifest.json"/>
```

#Before the closing </body> tag:

```
<script>
  if('serviceWorker' in navigator){
```

```
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>
offline.html:-
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8" />
  <title></title>
</head>
<body>
  <h2>offline</h2>
</body>
</html>
```

Output:

